

Multiple Choice (10 marks)

1. When the temperature of a strip of iron is increased, the length of the strip
 - (a) increases at first, then decreases.
 - (b) decreases in width as it gets longer.
 - (c) increases exponentially.
 - (d) remains the same.
 - (e) may increase and may decrease.
2. Using the speed of sound at sea level as approximately 750 miles per hour, what is the velocity of a jet plane traveling at Mach Number 2.2?
 - (a) 1900 mph
 - (b) 1750 mph
 - (c) 1320 mph
 - (d) 1200 mph
 - (e) 1800 mph
3. What is the radiant intensity of a Lambertian emitter which projects 50 mW into a hemisphere?
 - (a) .00995 (W /sr)
 - (b) .01095 (W/sr)
 - (c) .01295 (W/sr)
 - (d) .00695 (W /sr)
 - (e) .00295 (W/sr)
4. Assume that half of the mass of a 62-kg person consists of protons. If the half-life of the proton is 10^{33} years, calculate the number of proton decays per day from the body.
 - (a) 3.5×10^{-8}
 - (b) 4.8×10^{-5}
 - (c) 1.2×10^{-6}
 - (d) 6.3×10^{-4}
 - (e) 8.2×10^{-6}
5. When food is exposed to gamma radiation, the food

- (a) should be avoided
- (b) becomes slightly radioactive
- (c) will become carcinogenic
- (d) becomes inedible
- (e) will glow in the dark

True / False (10 marks)

Answer True or False

- 6. True or False: The current in the connecting wire of an LED lamp is half as much as the current flowing through the LED itself.
- 7. True or False: An object starting from rest with an acceleration of 16 ft/s^2 for 3 seconds will attain a speed of 48 ft/s.
- 8. True or False: An ocular lens with a focal length of 2.5 cm will yield an overall magnification of 100x when combined with an objective magnifying 10x.
- 9. True or False: When a voltage of 120 volts is applied to a light bulb with a resistance of 240 ohms, the current through the filament is 0.5 amperes.
- 10. True or False: The focal length of a diverging mirror is positive when the image is formed behind the mirror.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the process of alpha decay in uranium-238, including the resulting transformation of the nucleus and the implications for nuclear stability and radioactive decay chains.
- 12. Calculate the power density of a laser with a given power and beam diameter, considering a specific loss in the focusing system. Discuss the implications of your findings.

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